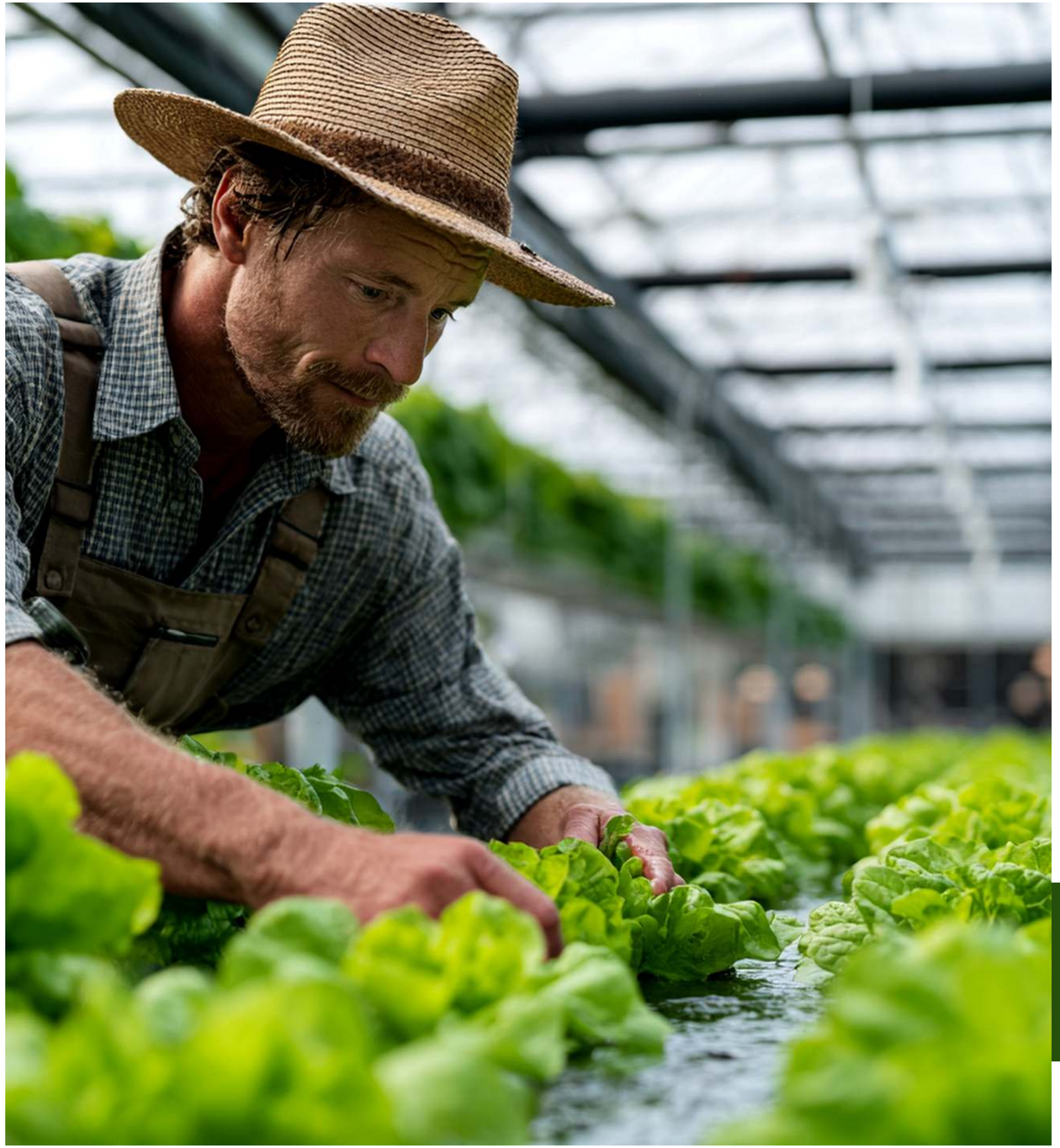
# AgriPredict

Innovating for a Sustainable Future

# AI





# Project Background

- Python-based prototype using machine learning for maize yield estimation
- Built on 7,258 Kenyan field observations (2016–2020)
- Inputs: location, elevation, soil, hybrid seed status
- Dashboard for yield estimates, annual patterns, calculator, and chatbot
- Supports agricultural planning for farmers and advisers

# Problem Definition

- Farmers must plan harvest resources without knowing exact yields
- Yield affected by soil, weather, and practices
- Analyzing records is time-consuming and technical
- AgriPredict AI offers experimental yield estimates and planning tools







# Project Objectives

- Develop and evaluate a working AI prototype
- Present clear yield estimates to support planning
- Define user needs and system requirements
- Prepare and validate dataset
- Visualize annual yield patterns
- Build and evaluate ML model
- Present results via dashboard and chatbot



# Business Success Criteria

- Accepts and validates complete field inputs
- Displays yield estimates and production scenarios
- Provides annual chart comparison
- Makes model error data accessible
- Answers supported questions and explains limitations













# Project Scope

## Included

- Kenyan maize field observations from the first growing season, 2016–2020.
- A Streamlit dashboard with an annual mean and median yield chart, a harvest calculator, and a chatbot for project questions, model results, and selected maize farming topics.

## Excluded

- Other crops, live weather and crop-price forecasts, and forecasts of future annual yields.
- Drone, sensor, and satellite integration, crop diagnosis, field-specific treatment advice, and automated farm control.
- Guaranteed harvest outcomes or financial returns.

# Tools & Techniques

- Data cleaning, multiple linear regression, bounded text assistant, and dashboard design.
- Python and its libraries, GitHub, and Canva.





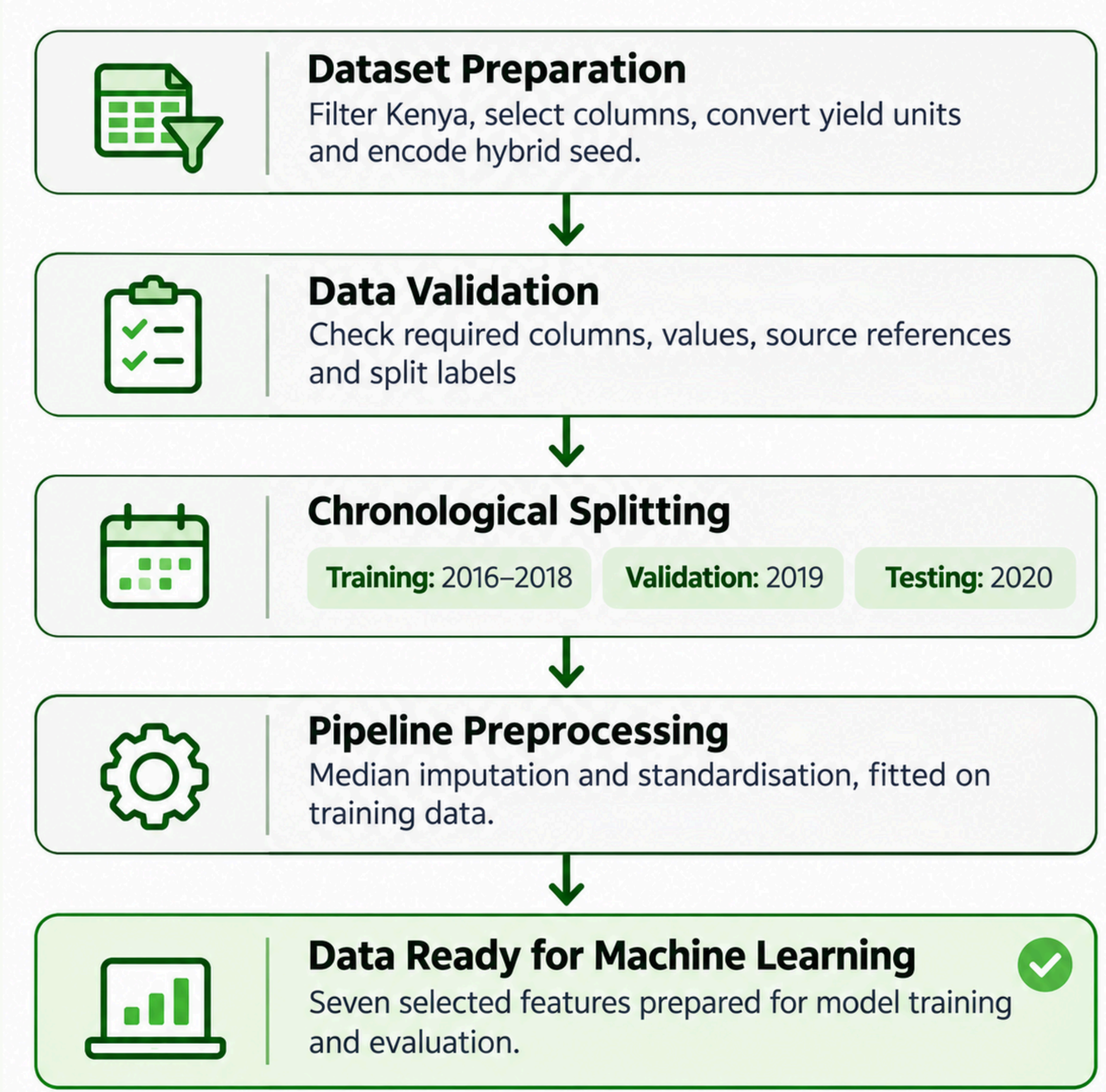





# Expected Business Benefits

- Improved harvest planning
- Clearer production scenarios
- Easy access to historical field info
- Transparent model evaluation
- Accessible explanations via chatbot assistant

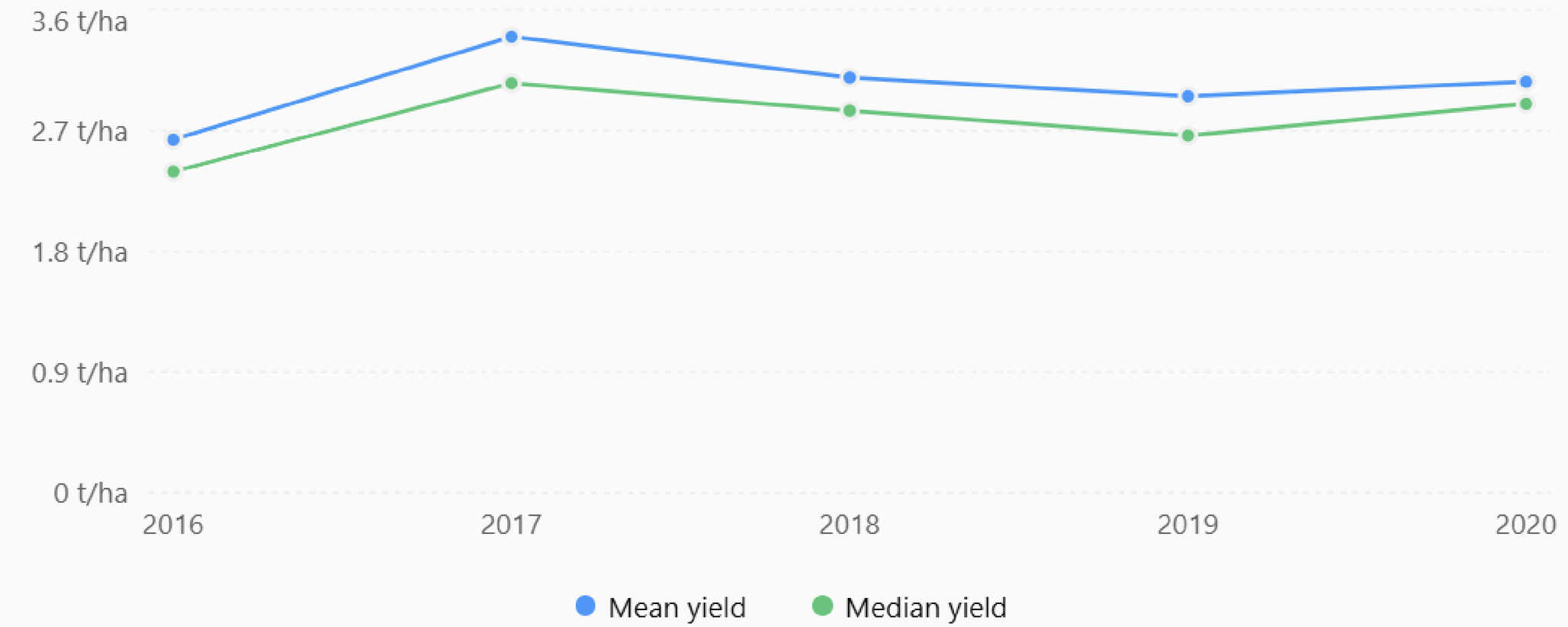
# Data Preparation



# Time Series Line Graph

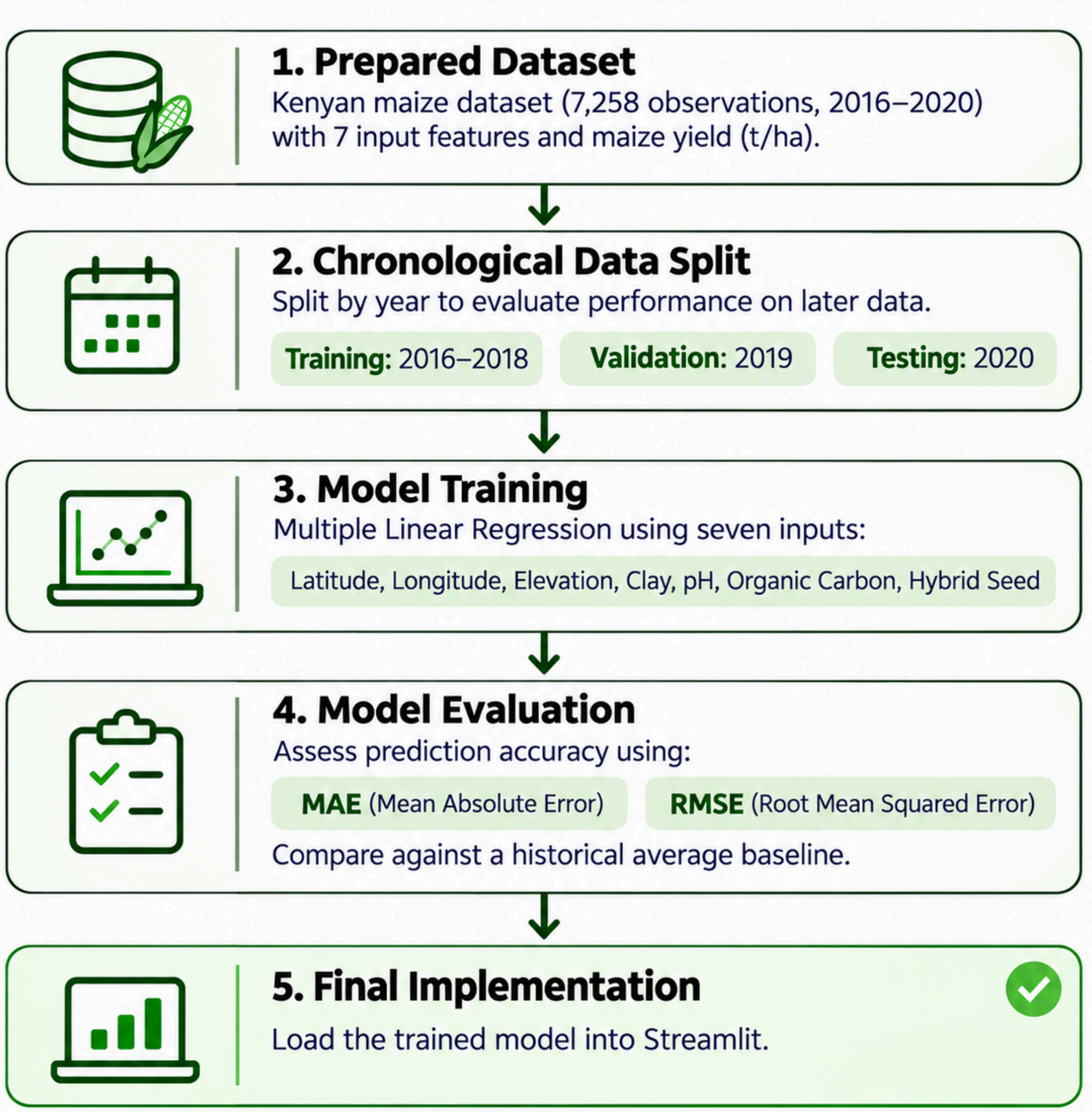
## Recorded maize yields in the Kenyan sample

Annual mean and median yield, 2016–2020

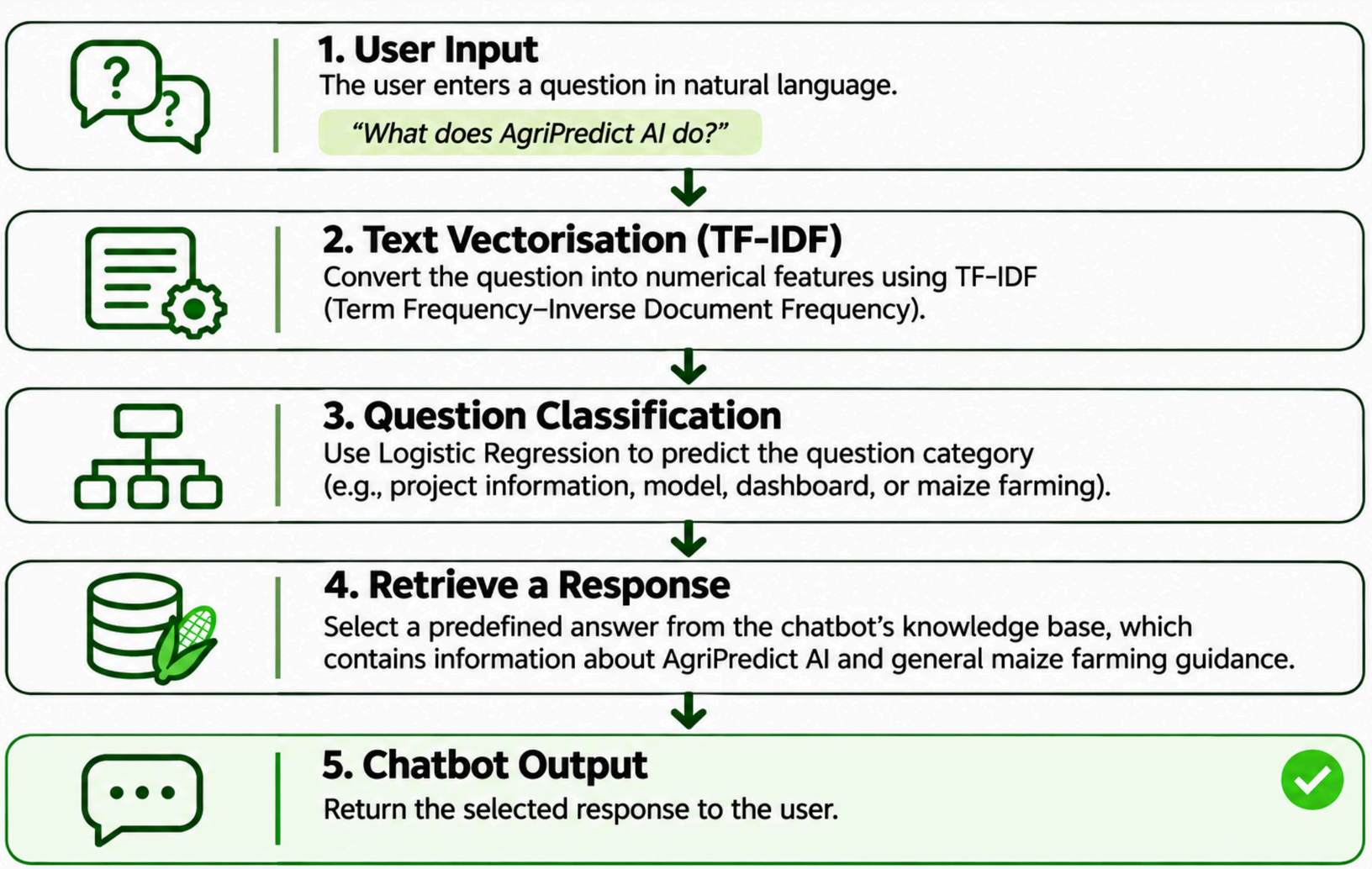



Source: AgriPredict AI annual yield analysis. Values in tonnes per hectare.

# Machine Learning



# NLP Chatbot


1. User Input
The user enters a question in natural language.
"What does AgriPredict AI do?"
2. Text Vectorisation (TF-IDF)
Convert the question into numerical features using TF-IDF (Term Frequency–Inverse Document Frequency).
3. Question Classification
Use Logistic Regression to predict the question category (e.g., project information, model, dashboard, or maize farming).
4. Retrieve a Response
Select a predefined answer from the chatbot's knowledge base, which contains information about AgriPredict AI and general maize farming guidance.
5. Chatbot Output
Return the selected response to the user.

# Dashboard

Open demo on GitHub



# Thank You